Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18RADE732]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

INTERNET OF THINGS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	List the different baseline technologies and write the comparison between IoT and M2M, IoT and WoT.	**08**
	b.	Describe MQTT methods and its components in brief. Mention its applications.	**06**
2.	**a.**	With a neat diagram, explain 6LoWPAN packet format and their addressing in connectivity technology. Mention its features.	**07**
	b.	Construct and explain the working principle of RFID. Mention its applications.	**07**
3.	**a.**	With the help of sketch, explain the different types in target tracking.	**07**
	b.	List and explain the different types of coverage problems in Static Wireless Sensor Networks.	**07**
		OR	
4.	**a.**	Sketch and explain the star and mesh Unmanned Air Vehicle Network Topology.	**08**
	b.	Discuss mid-end and high-end sensor nodes in Machine to Machine Communication.	**06**
5.	**a.**	Explain the following with respect to Arduino programming. i) Arrays ii) Interrupts	**06**
	b.	Sketch and explain the Servo Actuator using Arduino programming.	**08**
		OR	
6.	**a.**	Explain the data types in python and write a python program to generate CSV files using read and write operations.	**07**
	b.	Explain the different steps involved in socket programming using python.	**07**
7.	**a.**	List the components of Software defined network and discuss the rule placement and challenges for the same.	**07**
	b.	Discuss the sensor device and topology management in software defined WSN.	**07**
		OR	
8.	**a.**	Sketch the architecture of Mobile Traffic Management in software defined wireless management network and mention its key benefits.	**07**
	b.	Compare the different service models in cloud computing.	**07**